

Using Machine Learning to predict Hit Movies

Project by:

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Models used:

XGBoost, SVM, Logistic Regression, KNN, Random Forest,
Naive Bayes

The Dataset

The Dataset used in this project is the TMBd Movie Dataset from Kaggle:

- Total Rows = 10866
- Total Columns = 21

	popularity	budget	revenue	runtime	vote_count	vote_average	budget_adj	revenue_adj
count	10,866	10,866	10,866	10,866	10,866	10,866	10,866	10,866
mean	0.65	\$14,625,701.09	\$39,823,319.79	102 mins	217.39	5.97	\$17,551,039.82	\$51,364,363.25
std	1.00	\$30,913,213.83	\$117,003,486.58	31 mins	575.62	0.94	\$34,306,155.72	\$144,632,485.04
min	0.00	\$0.00	\$0.00	0 mins	10.00	1.50	\$0.00	\$0.00
25%	0.21	\$0.00	\$0.00	90 mins	17.00	5.40	\$0.00	\$0.00
50%	0.38	\$0.00	\$0.00	99 mins	38.00	6.00	\$0.00	\$0.00
75%	0.71	\$15,000,000.00	\$24,000,000.00	111 mins	145.75	6.60	\$20,853,251.08	\$33,697,095.72
max	32.99	\$425,000,000.00	\$2,781,505,847.00	900 mins	9,767.00	9.20	\$425,000,000.00	\$2,827,123,750.00

**The Descriptive Statistics of the dataset
(Before cleaning)**

RangeIndex: 10866 entries, 0 to 10865

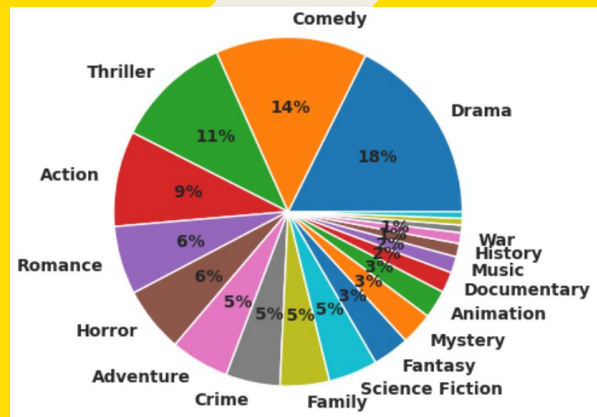
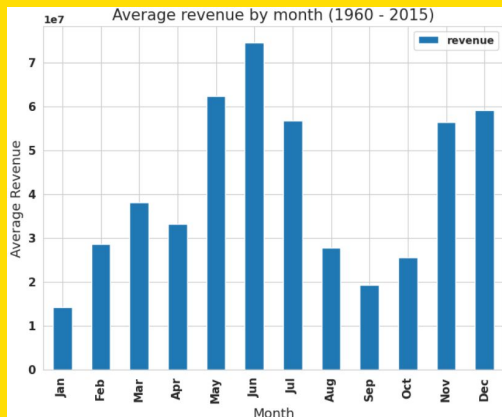
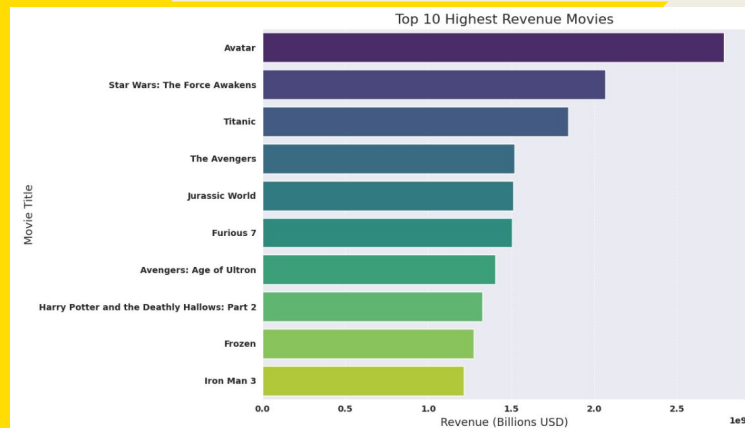
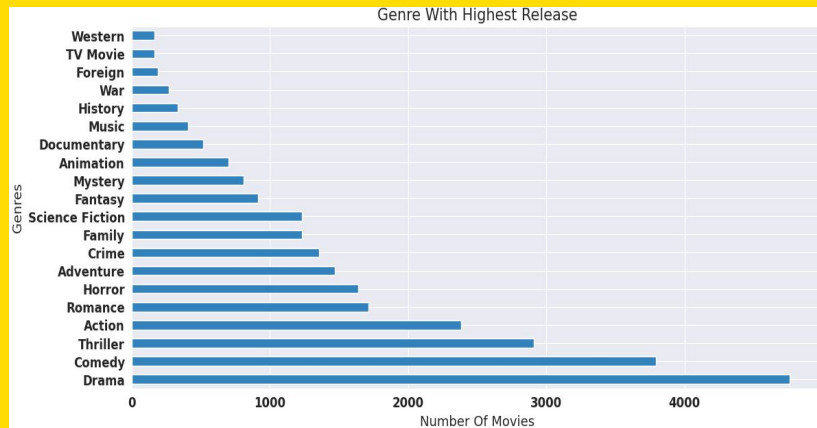
Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	id	10866 non-null	int64
1	imdb_id	10856 non-null	object
2	popularity	10866 non-null	float64
3	budget	10866 non-null	int64
4	revenue	10866 non-null	int64
5	original_title	10866 non-null	object
6	cast	10790 non-null	object
7	homepage	2936 non-null	object
8	director	10822 non-null	object
9	tagline	8042 non-null	object
10	keywords	9373 non-null	object
11	overview	10862 non-null	object
12	runtime	10866 non-null	int64
13	genres	10843 non-null	object
14	production_companies	9836 non-null	object
15	release_date	10866 non-null	object
16	vote_count	10866 non-null	int64
17	vote_average	10866 non-null	float64
18	release_year	10866 non-null	int64
19	budget_adj	10866 non-null	float64
20	revenue_adj	10866 non-null	float64

dtypes: float64(4), int64(6), object(11)

Original Column names and datatypes

Data Visualization



Data Preprocessing

In data preprocessing, we undertake the following procedures to clean the dataset and simplify the dataset:

- Removing duplicate rows
- Engineering the target feature (is_hit) and release_month feature from release_date
- Filtering rows so that budget, revenue > 0
- Filtering rows to keep rows with $60 \leq \text{runtime} \leq 240$
- Encoding movie genres with one-hot-encoding and dropping genres column
- Dropping pure text based columns to reduce noise
- Removing features that cause data leakage and overfitting

After processing the original dataset through these procedures, this is the final result:

```
Original Dataset shape: (10866, 21)
Processed Dataset shape: (3849, 28)
```

Dataset Balance

After processing the data we end up with a relatively balanced dataset:

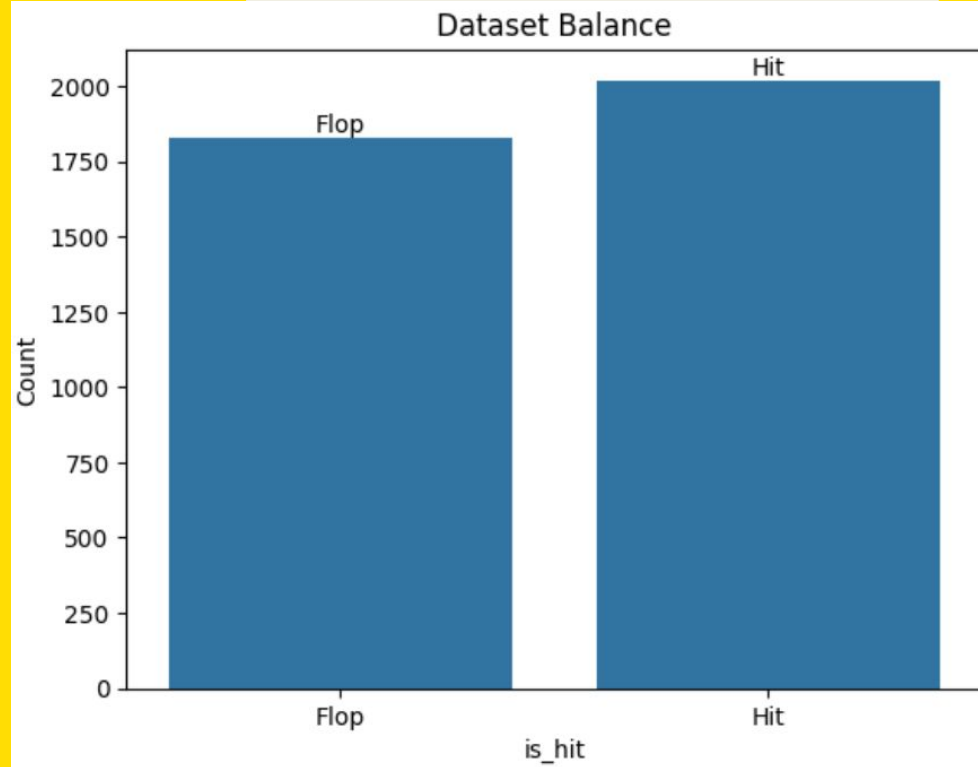
- Hit movies: 2020
- Flop Movies: 1829

Since the dataset is balanced we can use accuracy to evaluate our models.

We split the dataset into features (X) and target (y).

X = All columns // is_hit

y = is_hit



Train-Test Split and Feature Scaling

We split our dataframe into Train-Test with an 80:20 ratio.

Therefore we have this feature-target, Train-Test distribution:

```
Training features shape: (3079, 27)
Training target shape: (3079,)
Testing features shape: (770, 27)
Testing target shape: (770,)
```

- We have to scale our features to ensure each feature contributes equally to model training.
- We use standard scaling here as the dataset is balanced.
- We fit and transform the training data
- We only transform the testing to prevent overfitting.

Final Model Evaluations

Models	Baseline Accuracy	Enhanced Accuracy
SVM (4th)	68.57%	70.91%
Logistic Regression (5th)	70.64%	70.38%
KNN (6th)	62.70%	66.66%
Random Forest (1st)	72.71%	75.89%
XGBoost (2nd)	73.00%	74.40%
Naive Bayes (3rd)	69.12%	71.19%